

# From DPP Rate Equations to Finitary Birth Dynamics

Weak laws, common-noise realizations, and finite certificates on countable groups

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## Abstract

Fix a time-indexed determinantal process and a covariant pure-birth rate field that satisfies its finite-cylinder continuity equations. Several inequivalent realization questions arise: an ordinary weak law, an invariant weak law, a path determined by the supplied initial state and fresh Poisson noise, and a path whose individual coordinates admit finite certificates. We keep these notions separate and give sufficient conditions for the stronger levels.

Our main construction is a finite-rate-box criterion. It controls the worst-case oscillation of the rate over finite lower–upper boxes by an Osgood modulus and yields a total equivariant relative realization with an almost sure finite arbitrary-completion certificate for every coordinate path. A summable local-update hypothesis implies the rate-box criterion. For time-dependent mixture weights, this implication requires an explicit integrable factorization of the mixture tails. The direct local-update construction additionally gives first moments for the number and radius of relative queries.

Examples separate the criteria. A genuinely nonlocal diagonal-DPP rate field satisfies the rate-box condition but has no finite-mean local-neighborhood mixture. A static two-orbit example has a causal strong realization but no relative finite certificate. Finally, groups containing an infinite-order element admit a non-diagonal DPP family with a relative finitary realization and finite first query and radius moments in a constructed proper metric. These are sufficient theorems, not a solution of invariant weak existence for arbitrary Borel prescribed rates on nonamenable groups.

## 1 Introduction

A finite-cylinder forward equation specifies how the desired marginals should move, but it does not specify how much randomness or locality is needed to realize that movement. This distinction is particularly sharp for determinantal point processes (DPPs). The same curve of DPP marginals may be realized only as a weak martingale law, may admit a common-noise construction, or may admit a finitary algorithm whose answer at one coordinate is certified after finitely many input labels have been exposed.

This paper organizes these conclusions into a hierarchy. Let a countable group  $\Gamma$  act on a countable coordinate set  $I$ , let  $\mu_t$  be an invariant curve on  $\{0, 1\}^I$ , and let

$$a_i(t, x) = (1 - x_i)c_i(t, x)$$

be covariant pure-birth rates. The finite-cylinder continuity equation is

$$\mu_t(f) - \mu_0(f) = \int_0^t \mu_s \left( \sum_i a_i(s, x) [f(x^{i,1}) - f(x)] \right) ds \quad (1)$$

for every cylinder function  $f$ . Equation (1) is necessary for a prescribed-marginal dynamics, but it does not imply invariance, pathwise uniqueness, common-noise measurability, or finite certificates.

The main result starts from an ordinary weak realization and adds a geometric condition on the rate field. Expose only finitely many coordinates, allow all outside completions, and compute the exact lower and upper possible rates. If the expected gap of these two thresholds closes with an Osgood modulus, the lower and upper graphical envelope processes coalesce. Their eventual agreement produces a finite certificate for the complete coordinate path.

The second contribution compares this criterion with a more concrete summable local-update condition. The comparison is not merely formal when the mixture weights depend on time: a tail that is small pointwise in time is not automatically bounded by a fixed deterministic sequence. We supply the required  $L^1$  tail factorization and obtain a linear Osgood modulus.

**Scope.** The results do not prove arbitrary-group invariant weak existence, a general strong factor construction for unbounded or infinite-range rates, or exact ordered joint iid sampling for arbitrary DPP kernels. These remain distinct open problems.

## 2 Five realization levels

The following table records the outputs used throughout the paper.

| Level                    | Output                                                                                              | Not included automatically                         |
|--------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Ordinary weak            | A martingale law with marginals $\mu_t$ and rates $a$                                               | Invariance, common noise, uniqueness               |
| Invariant weak           | An invariant path law with the same properties                                                      | An iid or Poisson construction                     |
| Relative strong          | An equivariant measurable path from supplied $X_0$ and fresh product noise                          | Finite certificates                                |
| Relative finitary        | Each coordinate path is fixed by finitely many complete input labels under every outside completion | Query moments                                      |
| Unconditional iid factor | First sample $X_0$ equivariantly from iid, then apply a relative construction                       | Finitary or moment bounds unless proved separately |

Weak convergence of joint laws does not preserve a common-input factor representation. Likewise, separate iid samplers for two marginals do not produce an ordered joint factor. Each arrow in the hierarchy therefore needs its own argument.

## 3 Common setup and weak inputs

For clarity, first take  $I = \Gamma$  with the regular action. Write  $x^{g,b}$  for the configuration obtained by setting coordinate  $g$  to  $b$ . Intrinsic rates  $c_g$  are independent of the center bit and actual rates are

$$a_g(t, x) = (1 - x_g)c_g(t, x).$$

Every pure-birth rate that vanishes at occupied sites has this form after setting  $c_g(t, x) = a_g(t, x^{g,0})$ .

The ordinary weak-existence input assumes a weakly continuous prescribed curve, finite-valued jointly Borel rates, equation (1), and

$$\int_0^1 \mu_t(a_i(t, \cdot)) dt < \infty$$

for every coordinate. It yields a coordinatewise cadlag pure-birth martingale law with the prescribed marginals and no simultaneous positive-time births. In the linear DPP application one takes  $K_t = C + tH$ , with  $H \succeq 0$  and

$$\varepsilon I \preceq K_t \preceq (1 - \varepsilon)I.$$

For bounded rates, a weak law in this pure-birth, no-simultaneous-birth class admits a compatible enlargement by independent sitewise Poisson random measures. Compatibility means that future product-noise increments are independent of the common past and that the path accepts exactly the points below its current rate.

For countable amenable groups, averaging the compact convex solution set gives an invariant prescribed-marginal weak law. In the stated linear DPP setting, for arbitrary countable groups, invariant weak existence is equivalent to a local endpoint condition along a vanishing-mesh sequence of partitions: every adjacent time interval must admit a diagonally invariant dynamically realizable endpoint law. Existence of these local invariant endpoint elements remains open in general.

## 4 Finite rate boxes

Let  $F_1 \subset F_2 \subset \dots$  be finite sets containing the identity. Given lower and upper patterns  $l \leq u$  on  $gF_R$ , let

$$B_g^R(l, u) = \{z : l_j \leq z_j \leq u_j \text{ for } j \in gF_R\}.$$

Define exact intrinsic extrema

$$\ell_g^R(t, l, u) = \inf_{z \in B_g^R(l, u)} c_g(t, z), \quad h_g^R(t, l, u) = \sup_{z \in B_g^R(l, u)} c_g(t, z). \quad (2)$$

**Definition 4.1** (BOX). Both extrema in (2) are jointly Borel in time and the finite patterns.

Drive lower and upper finite-range envelope processes by the same proposal field using  $\ell^R$  and  $h^R$ . Let

$$\delta_R(t) = \mathbb{P}(L_e^R(t) \neq U_e^R(t))$$

and let  $W_e^R(t)$  be the current threshold gap.

**Definition 4.2** (WIDTH). There exist constants  $A_n, \tau_n \geq 0$ , with  $\tau_n \rightarrow 0$ , and  $L \in L^1[0, 1]$  such that, outside one null time set,

$$\mathbb{E}W_e^R(t) \leq L(t)(A_n \delta_R(t) + \tau_n), \quad R \geq n. \quad (3)$$

Put

$$\omega(r) = \inf_n (A_n r + \tau_n).$$

We impose either  $\omega \equiv 0$ , or  $\omega(r) > 0$  for  $r > 0$  and

$$\int_{0+} \frac{dr}{\omega(r)} = \infty. \quad (4)$$

**Theorem 4.3** (Shrinking rate boxes). *Assume ordinary weak existence, a uniform finite rate bound  $M$ , compatible Poisson completion, BOX, WIDTH, and (4). Then there is a total Borel exactly equivariant relative realization of the prescribed process. Almost surely, every complete coordinate path has a finite arbitrary-completion certificate. The prescribed-marginal weak law is unique in the stated class and invariant. If the initial law has a finitary iid sampler with the same certificate convention, then the full path is an iid finitary process after composition.*

*Proof roadmap.* At fixed  $R$ , the backward ancestor graph is finite because strict-time chains of length  $k$  have expected count at most  $(M|F_R|)^k/k!$ . Exact extrema make the envelope systems nested in  $R$ , while compatible Poisson completion traps every weak reference path between them. Compensation gives

$$\delta_R(t) \leq \int_0^t \mathbb{E} W_e^R(s) ds.$$

After taking  $R \rightarrow \infty$  in the correct order, (3)–(4) and Osgood’s lemma force the limiting discrepancy to vanish. Each coordinate chooses its birth time from a fixed finite list, so agreement at all times upgrades to eventual equality of the entire coordinate paths at some finite radius. The least successful local computation supplies the arbitrary-completion certificate.  $\square$

The theorem asserts no query-size or radius moment, no simultaneous all-initial collapse event, and no result for arbitrary unbounded rates.

## 5 Summable local updates imply shrinking boxes

**Definition 5.1** (LOCAL). Suppose the actual rates have the decomposition

$$a_g(t, x) = \sum_{k \geq 0} \lambda_k(t) b_k(t, (x_{gv})_{v \in V_k}), \quad (5)$$

where  $V_k$  is finite and contains the identity,  $\lambda_k : [0, 1] \rightarrow [0, \infty)$  is Borel,  $b_k : [0, 1] \times \{0, 1\}^{V_k} \rightarrow [0, 1]$  is Borel and vanishes when the center is occupied, and

$$\sum_k \lambda_k(t) |V_k| \leq B < \infty \quad \text{for every } t. \quad (6)$$

**Theorem 5.2** (LOCAL implies BOX+WIDTH). *Under (5)–(6), the center-independent intrinsic rate satisfies BOX and WIDTH for a nested sequence of finite sets. One may take  $A_n = B$ ,  $\tau_n = 2^{-n}$ , and an integrable  $L \geq 1$ . Consequently  $\omega(r) = Br$ , with the zero case understood separately.*

*Proof.* Replace each component by its center-vacant version. Then

$$c_e(t, x) = \sum_k \lambda_k(t) \tilde{b}_k(t, x|_{V_k}).$$

For fixed time the series converges uniformly in  $x$  because  $\sum_k \lambda_k(t) \leq B$ . Thus  $c_e(t, \cdot)$  is continuous. On every finite cylinder, exact extrema may be computed over a countable dense family of extensions, so they are Borel in time. Hence BOX holds.

Put

$$s_N(t) = \sum_{k > N} \lambda_k(t).$$

Then  $0 \leq s_N \leq B$  and  $s_N(t) \downarrow 0$ . Dominated convergence permits a strictly increasing sequence  $N_n$  such that

$$\int_0^1 s_{N_n}(t) dt \leq 2^{-2n}.$$

Set

$$L(t) = 1 + \sum_{n \geq 1} 2^n s_{N_n}(t). \quad (7)$$

Then  $L \in L^1$ ,  $L \geq 1$ , and  $s_{N_n}(t) \leq 2^{-n} L(t)$  outside one null set. Let

$$F_n = \bigcup_{k \leq N_n} V_k.$$

These sets are finite, nested, and contain the identity.

For  $R \geq n$ , a component  $k \leq N_n$  contributes no width if the two left-limit envelope configurations agree on  $V_k$ , and otherwise contributes at most one. The tail contributes at most  $s_{N_n}(t)$ . Translation invariance and a union bound give

$$\begin{aligned} \mathbb{E} W_e^R(t) &\leq \sum_{k \leq N_n} \lambda_k(t) |V_k| \delta_R(t) + s_{N_n}(t) \\ &\leq B \delta_R(t) + 2^{-n} L(t) \\ &\leq L(t) (B \delta_R(t) + 2^{-n}). \end{aligned}$$

This is WIDTH. Finally

$$\omega(r) = \inf_n (Br + 2^{-n}) = Br,$$

which satisfies the Osgood condition when  $B > 0$ ; if  $B = 0$ , all rates vanish.  $\square$

The factorization (7) is essential at this level of generality. The simpler choice  $\tau_n = \sum_{k > n} \lambda_k$  is valid for time-independent weights but is not a deterministic WIDTH constant for arbitrary  $\lambda_k(t)$ .

## 5.1 Quantitative addendum

The direct LOCAL graphical construction gives more than the qualitative conclusion of the rate-box theorem. If the weights are constant and

$$B_1 = \sum_k \lambda_k \sum_{v \in V_k} |v| < \infty$$

in a fixed left-invariant metric, then the expected number of relative queried labels is at most  $e^B$ , and the expected maximum query radius is at most  $B_1 e^B$ . These moments need not survive composition with a separate initial iid sampler.

## 6 Mean Osgood uniqueness

Rate boxes control every outside completion. A weaker, nonfinitary conclusion follows from a condition involving only couplings of the prescribed marginals. Assume the covariant rates are bounded, an ordinary prescribed-marginal weak law exists in the no-simultaneous-birth class,  $L \in L^1[0, 1]$ , and  $\omega$  is continuous, nondecreasing, vanishes at zero, is positive away from zero, and

satisfies (4). Suppose that, outside one fixed null set of times, every coupling  $(U, V)$  with both laws equal to  $\mu_t$  satisfies

$$\sup_i \mathbb{E}|a_i(t, U) - a_i(t, V)| \leq L(t) \omega \left( \sup_j \mathbb{P}(U_j \neq V_j) \right). \quad (8)$$

**Theorem 6.1** (Mean Osgood extraction). *Under the preceding assumptions, compatible conditional copies driven by the same Poisson field agree. The conditional law given the initial state and noise is almost surely Dirac. Hence the prescribed weak law is unique within the prescribed- $\mu_t$  martingale class, invariant, and has a causal relative strong realization.*

This theorem gives no finite certificate and is not an all-initial-law uniqueness theorem.

## 7 Examples and separations

### 7.1 A nonlocal rate-box example

For every countably infinite group  $G$ , take  $\Gamma = G \times C_2$ , pair each site with its  $C_2$  partner, and use a lexicographic comparison of two enumerated rays to perturb opposite rates. The perturbations cancel in every cylinder equation, so the prescribed marginals remain iid Bernoulli and hence diagonal DPPs. Finite prefixes give

$$\mathbb{E}W_e^R(t) \leq \left( \frac{1}{6} + \frac{5n}{12} \right) \delta_R(t) + \frac{5}{24} \left( \frac{17}{32} \right)^n.$$

The induced modulus obeys

$$\frac{7}{12}r \leq \omega(r) \leq 2r(1 + \log(1/r)),$$

so the rate-box theorem gives a relative whole-path finitary realization. Every off-root coordinate nevertheless has positive essential influence on the root rate, and the summed worst-case coordinate oscillation is infinite. Thus no finite-mean nonnegative LOCAL mixture exists.

### 7.2 Mean control does not imply finite certificates

On a two-orbit  $(\mathbb{Z} \times C_2)$ -set, retain a static iid environment and let active-pair rates depend on a closed tail event with empty interior. Conditional on the environment, independent pair clocks give a causal equivariant strong process with diagonal-DPP marginals and satisfying (8). The generator nevertheless forces a discontinuous conditional root-endpoint kernel on a positive-measure set. A relative arbitrary-completion finite certificate would make an everywhere-defined version continuous at almost every input, a contradiction.

### 7.3 Failure of the box method is not failure of joint finitariness

A single-orbit dynamic replacement has exact DPP marginals and a causal relative strong realization. Its finite-box widths stay bounded away from zero on an event of positive probability, so the exact-box criterion fails for every nested query family. Its one-root endpoint conditional law is nevertheless almost-everywhere continuous and that single finite-valued output has a relative finite sampler. This does not produce a finitary realization of the complete joint process: independently sampling the one-coordinate kernels would destroy the joint law.

## 8 A non-diagonal DPP application

Assume  $\Gamma$  contains an element of infinite order. A positive sandwich construction gives a full-support convolution probability  $v = a * w * a$  and a positive contraction  $V = T_v = AWA$ , with  $\ker V = \ker A$ , where  $A$  is a cyclic spectral-zero operator. For a weighted-small finite-support Hermitian background  $b$ , set

$$K_t = \frac{1}{2}I + T_b + \delta t V^2.$$

This kernel curve is uniformly gapped. Its exact DPP birth rates satisfy the cylinder equations and admit a summable local-update decomposition after the inverse-pair coefficients of the convolution kernel are ordered.

**Theorem 8.1** (Cyclic non-diagonal application). *For the preceding family, there is a total equivariant relative finitary DPP process. In a constructed proper left-invariant metric, both the relative query count and query radius have finite first moments. Moreover, if a bounded finite-propagation operator  $T$  satisfies  $TT^* \preceq V^2$ , then  $T = 0$ .*

Thus the increment has no nonzero finite-propagation square minorant. A separate gapped initial sampler may be composed with the relative process to obtain a joint iid finitary endpoint pair, but this composition does not transfer the relative moment bounds. The construction requires an infinite-order element, a constructed increment, weighted-small backgrounds, and a constructed metric; it is not general exact ordered joint iid sampling.

## 9 Logical picture and open problems

The proved implications can be summarized as

$$\text{finite range} \subsetneq \text{LOCAL} \subsetneq \text{BOX+WIDTH} \subseteq \text{relative finitary} \subseteq \text{relative strong} \subseteq \text{weak}.$$

The first two strict containments are witnessed by the cyclic and nonlocal rate-box examples. The static environment separates relative strong from relative finitary. Necessity of BOX+WIDTH for full-process finitariness is not known.

We close with four open problems.

- (1) For arbitrary nonamenable groups, do all finite-valued Borel covariant DPP birth rates satisfying the integrated cylinder equations and ordinary weak-existence hypotheses admit an invariant weak realization?
- (2) Which rates outside BOX+WIDTH admit a relative finitary full-process realization? The dynamic example settles only a one-output kernel.
- (3) Can arbitrary ordered DPP kernels be joined by one exact equivariant iid factor?
- (4) Can the cyclic application's query moments be obtained in a preassigned word metric or after composition with the initial sampler?

## Draft status and provenance

This document is a theorem-level working draft assembled from separately audited internal research notes. The comparison  $\text{LOCAL} \Rightarrow \text{BOX+WIDTH}$  has received a fresh internal scope audit. Novelty, external domain review, formal verification, and journal acceptability have not been certified. Full source attributions and proof appendices will be added before any submission version.